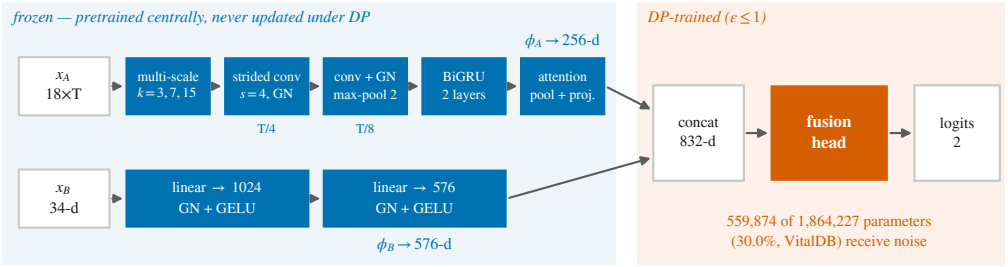
frozen — pretrained centrally, never updated under DP
DP-trained (ε ≤ 1)
x_A
18×T
multi-scale
$k = 3, 7, 15$
strided conv
$s = 4$, GN
T/4
conv + GN
max-pool 2
T/8
BiGRU
2 layers
attention
pool + proj.
$\phi_A \rightarrow$ 256-d
x_B
34-d
linear → 1024
GN + GELU
linear → 576
GN + GELU
$\phi_B \rightarrow$ 576-d
concat
832-d
fusion
head
logits
2
559,874 of 1,864,227 parameters
(30.0%, VitalDB) receive noise